

Chapter 1

EXERCISE 1. Classify the following equations as ordinary or partial, give their order, and state whether they are linear or nonlinear. In each case identify the dependent and independent variables.

- (i) Bessel's equation (ν is a parameter) $x^2 y'' + xy' + (x^2 - \nu^2)y = 0$,
- (ii) Burgers' equation (ν is a parameter) $\frac{\partial u}{\partial t} - \nu \frac{\partial^2 u}{\partial x^2} + u \frac{\partial u}{\partial x} = 0$,
- (iii) van der Pol's equation (m, k, a , and b are parameters) $m\ddot{x} + kx = a\dot{x} - b\dot{x}^3$,
- (iv) $\frac{dy}{dt} = t - y^2$,
- (v) the wave equation (c is a parameter) $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$,
- (vi) Newton's law of cooling (k is a parameter and $T_A(t)$ is a specified function) $\frac{dT}{dt} = -k(T - T_A(t))$,
- (vii) the logistic population model (k is a parameter) $\frac{dp}{dt} = kp(1 - p)$,
- (viii) Newton's second law for a particle of mass m moving in a potential $V(x)$, $m\ddot{x} = -V'(x)$,
- (ix) the coupled equations
$$\begin{aligned}\dot{x} &= x(4 - 2x - y) \\ \dot{y} &= y(9 - 3x - 3y),\end{aligned}$$
- (x) $\frac{d\mathbf{x}}{dt} = A\mathbf{x}$, where $\mathbf{x}$ is an n -component vector and A is an $n \times n$ matrix.

EXERCISE 2. Let $x(t)$ be the population of a certain species as a function of time, and suppose that the rate of change of this population is proportional to the existing population at each time, with constant of proportionality $k \in \mathbb{R}$.

- (i) Write a differential equation for $x(t)$.
- (ii) Solve it, determine the behavior of $x(t)$ as $t \rightarrow \infty$ and interpret it depending on the sign of k .
- (iii) If the initial population is x_0 and $k > 0$, determine the time needed for the population to double.

EXERCISE 3. Plutonium 239, virtually non-existent in nature, is one of the radioactive materials used in the production of nuclear weapons, and is a by-product of the generation of power in a nuclear reactor. Its half-life (i.e., the time it takes for half of the radioactive isotopes to decay) is approximately 24 000 years. Taking $dN/dt = -kN$ as an appropriate model for radioactive decay, where $N = N(t)$ is the number of isotopes at time t , what is the value of k that should be used for this isotope?

EXERCISE 4. A cup of hot chocolate is initially at temperature T_0 and is left in the room with an ambient temperature $T_A(t)$. The rate of cooling of the hot chocolate is proportional to the difference between the current temperature $T(t)$ and the ambient temperature.

- (i) Which are the dependent variable, the independent variable, and the parameters?
- (ii) Write a linear differential equation for the temperature of the hot chocolate that takes into account heat exchange with the room. Such a differential equation is known as Newton's law of cooling.

EXERCISE 5. A body is found in a cold room ($T_A = 5^\circ C$) at 3 p.m. and its temperature then is $19^\circ C$. An hour later its temperature has dropped to $15^\circ C$. Use Newton's law of cooling to estimate the time of death, assuming that the (live) body temperature is $37^\circ C$ and measuring time in hours.

EXERCISE 6. A tank contains 100 litres of water with 1 kg. of salt dissolved in it (uniformly). At time $t_0 = 0$, water without salt flows into the tank at a rate of 1 litres per minute, while the mixture flows out at the same rate. Write a differential equation for $x(t)$, which is the number of kg. of salt per litre of water dissolved at time $t > 0$. How much salt is dissolved in the tank as $t \rightarrow \infty$? What happens if water with s kg. of salt per litre flows into the tank at a given rate while the mixture (that is kept uniform) flows out at the same rate?

EXERCISE 7. Consider an object in free fall from an initial point y_0 and with initial velocity 0, under the action of gravity and neglecting air friction. Write a differential equation for the position of the object $y(t)$ and solve it.

EXERCISE 8. Consider a simple pendulum consisting of a bob of mass m at the end of a string of length ℓ and negligible mass, and denote by $\theta(t)$ the angle with respect to the vertical as function of time t . Neglecting friction, apply Newton's second law to write a differential equation for $\theta(t)$.

EXERCISE 9. A refinement over Malthus' population growth model discussed in Chapter 1 is provided by the logistic equation:

$$\frac{dp}{dt} = kp \left(1 - \frac{p}{M}\right),$$

where $p(t)$ is the population size at time t , k is its growth rate (difference between birthrate and deathrate), and M is the maximum sustainable population. Solve the equation for a generic initial condition $p(t_0) = p_0$ and discuss the evolution of the population for $t \rightarrow \infty$, comparing with Malthus' ($M \rightarrow \infty$) prediction.

EXERCISE 10. In general, it is not possible to find explicit solutions of the Navier-Stokes equations that govern fluid flow. However, in certain cases the equations simplify very strongly. Suppose that a fluid is flowing down a pipe that has a circular cross-section of radius a . Assuming that the velocity V of the fluid depends only on its distance from the centre of the pipe, the equation satisfied by V is

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dV}{dr} \right) = -P,$$

where $P > 0$. Multiply by r and integrate once to show that

$$\frac{dV}{dr} = -\frac{Pr}{2} + \frac{c}{r},$$

where c is an arbitrary constant. Integrate again to find an expression for the velocity, and then use the facts that (i) the velocity should be finite at all points in the pipe and (ii) that fluids 'stick' to boundaries (which means that $V(a) = 0$) to show that

$$V(r) = \frac{P}{4}(a^2 - r^2). \quad \text{"Poiseuille flow"}$$
